Models	Acc. (%) ↑
Continue pre-training	0.5
DoT-finetune	32.6
(-) scheduled sampling	31.2
DoT^{MP}-finetune	37.7
(-) coupled sampling	35.5

Table 2: Ablation of Plaid DoT on GSM8K.

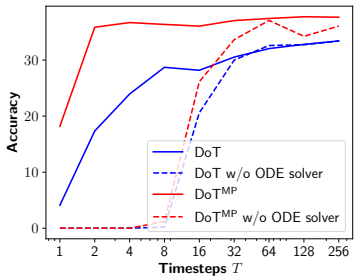


Figure 3: The effectiveness of ODE solver in speedup inference of Plaid DoT.

4.3 Results on Grade School Math

We now move on to a much more complex grade school math task GSM8K as shown in the right part of Table 1. We first consider training DoT from scratch as in the previous tasks, but we are only able to achieve an accuracy of around 5%, which is much lower than the fine-tuned version of GPT-2. This indicates the pre-trained natural language understanding capability is vital for grade school math. Once DoT is extended based on the pre-trained diffusion language models Plaid and SEDD, the performance is significantly improved after fine-tuning, where the DoT based on SEDD-medium outperforms similar-sized GPT2-medium with CoT by around 10%. Additionally, multi-pass DoT, with casual bias, performs slightly better than single-pass one on Plaid, while the latter is more efficient. The performance gap between SEDD and Plaid also highlights the importance of the training objective in pretraining diffusion LMs. Finally, we find that self-consistency further yields substantial improvements in DoT models owing to the diverse generations of diffusion model (§4.5).

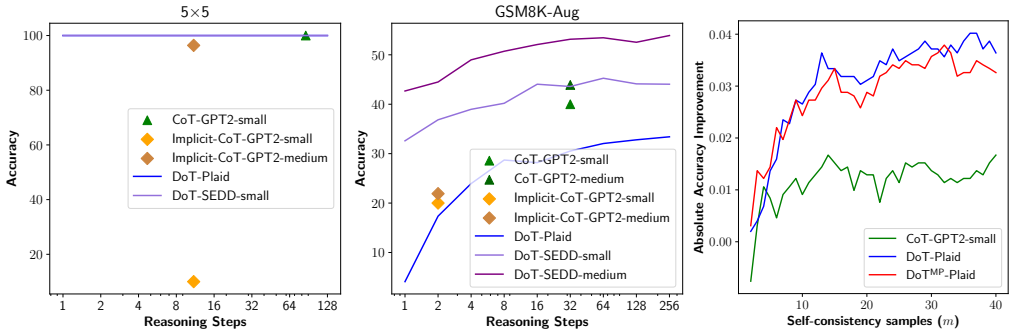
We further explore several alternatives and conduct an ablation study as in Table 2 when fine-tuning Plaid. As discussed above, continuing pre-training Plaid using the GSM8K-augmented dataset and performing reasoning with gradient-based conditioning is not a good choice for fine-tuning diffusion LMs on downstream tasks, because reasoning tasks require more specific guidance. An example of groundtruth and recovered text is shown below, where bold words in the query part are incorrectly recovered:

> *Groundtruth*: Two trains leave San Rafael at the same time. They begin traveling westward, both traveling for 80 miles. The next day, they travel northwards, covering 150 miles. What's the distance covered by each train in the two days? «2∗80=160» «150∗2=300» «300+160=460» «460/2=230» #### 230
>
> *Recovered Text*: **Three** trains leave San Juan at the same time. They **start** traveling westward, both traveling for 80 miles. The next day, they travel **southward**, covering 150 miles. What's the distance covered by each train in the two days? «3∗80=180» «180+80+150=340» «340/30=12.5» #### 12.5

We can see there are three recovered query tokens that exhibit minor differences due to soft gradient guidance, causing interference with the model's comprehension of the problem. The ablation of two sampling strategies proposed in §3.3 showcases their effectiveness. This provides evidence that better denoising models are trained using our training-time sampling strategies, allowing DoT models to self-correct more effectively during inference. Further analysis about self-correction is listed in §4.6. In Figure 3, we further show the conditional ODE solver substantially speeds up the inference of continuous diffusion model Plaid, ensuring a decent performance with only 8 generation timesteps.

4.4 Reasonability-efficiency Trade-off

The community has devoted substantial efforts to improve the reasoning capabilities of left-to-right language models, such as refining instructions [31, 67], finding better demonstrations [9, 55, 58], and designing elaborate decoding algorithm [43, 56, 57]. Non-autoregressive diffusion models naturally provide another simple way to enhance reasoning by allocating more timesteps during inference, albeit at the expense of efficiency. We show such efficiency trade-off in Figure 4(a), where we



(a) (b)

Figure 4: **(a)** Accuracy over reasoning steps using various methods. We measure the reasoning steps as the average number of calling the model `forward` function for instances from the test set. DoT provides a flexible way to balance accuracy and efficiency through the reasoning steps. **(b)** Absolute accuracy improvement versus samples in self-consistency per instance on the GSM8K dataset with Plaid DoT.

measure the reasoning steps as the average number of calling the model `forward` function for all the instances from the test set. For CoT and Implicit CoT baselines, we treat reasoning steps as the average number of output tokens for all the test instances[5].

Given a small budget of reasoning steps (e.g., 1 or 2) on simpler tasks such as 5×5, both DoT-Plaid and DoT-SEDD already have an accuracy of 100%, and no more reasoning steps are needed. For such cases of simple tasks, only a little computation cost is required for our method. For complex tasks such as GSM8K, we find DoT performance can continuously improve by allowing more reasoning steps, which indicates DoT can be efficient if we can sacrifice performance in certain scenarios. Specifically, DoT-SEDD-medium outperforms autoregressive CoT-GPT2-medium when we allocate 32 generation timesteps, and the performance continues improving when we increase the timesteps. In comparison, CoT and Implicit CoT with the autoregressive model are hard to be more efficient given their nature of token-by-token prediction. Overall, with DoT, we can flexibly control the trade-off between efficiency and performance for tasks with different difficulty levels.

4.5 Self-consistency in DoT

Figure 4(b) shows the effectiveness of the self-consistency mechanism for Plaid DoT and its variant. We can see self-consistency improves both DoT and DoT^{MP}, which is in line with the effectiveness of self-consistency for auto-regressive models [52]. From Table 1, SEDD DoT is also significantly improved by self-consistency. This benefits from the diversity generation in DoT. We observe that DoT can generate diverse reasoning paths, such as `<3*3=9><9*60=540>` and `<3*60=180><180*3=540>` for the same question, providing cross-validation when selecting the most "consistent" answer. Note that different from autoregressive models, where diversity usually relies on decoding algorithms [8, 22], the natural advantage of the diffusion models is to generate different sentences with different random noises at each timestep.

4.6 Self-correction in DoT

In this section, we provide several cases in Table 3 to show the self-correction ability of Plaid DoT, which acts as a distinct difference between diffusion models and autoregressive models. In the first case, we can see the model figures out all the correct thoughts together with only a single reasoning step (i.e., a single calling of the model `forward` function), and obtains the correct final answer in the second step. This mirrors how humans think in both fast and slow modes [26]. In the second case where the problem is slightly harder, the model cannot give concrete thoughts in the first step but can still produce the correct answer through the later "slow" thinking process. We can see the solution framework, roughly outlining how the task will be carried out, is established at the very

[5]Here we define reasoning steps of Implicit CoT as the times of forwarding the whole model instead of the layers of transformers, considering that the former reflects the inference speed.

Table 3: Cases that show the predictions of Plaid DoT at each time-step t with T=8 on the GSM8K test set. The incorrect thoughts are marked in bold red and we omit some correct predictions when $t < 4$. The difficulty level of the questions increases from left to right.

Q.	A robe takes 2 bolts of blue appet and half that much white fertil. How many bolts in total does it take?	Tommy is fundraising for his charity by selling brownies for $3 a slice and cheesecakes for $4 a slice. If Tommy sells 43 brownies and 23 slices of cheesecake, how much money does Tommy raise?	When Freda cooks canned tomatoes into sauce, they lose half their volume. Each 16 ounce can of tomatoes that she uses contains three tomatoes. Freda's last batch of tomato sauce made 32 ounces of sauce. How many tomatoes did Freda use?
Gold	<<2/2=1>> <<2+1=3>> #### 3	<<43*3=129>> <<23*4=92>> <<129+92=221>> #### 221	<<32*2=64>> <<64/16=4>> <<3*4=12>> #### 12
***t*=8**	<<2/2=1>> <<2+1=3>> **#### 43**	**<<43*3=123>> <<3*43=12>> <<133+12=217>> # # 227**	<<32/16=2>> **<<12/12=2*3 ## *2=2= ######**
***t*=7**	<<2/2=1>> <<2+1=3>> #### 3	<<43*3=129>> <<23*4=92>> **<<129+92=111>> #### 111**	<<32/16=2>> **#2*3=4>> #4*3=3>>>>>>#######**
***t*=6**	<<2/2=1>> <<2+1=3>> #### 3	<<43*3=129>> <<23*4=92>> <<129+92=221>> #### 221	<<32/16=2>> **<<2*3=6>> <<6*3=24>> #### 18**
***t*=5**	<<2/2=1>> <<2+1=3>> #### 3	<<43*3=129>> <<23*4=92>> <<129+92=221>> #### 221	<<32/16=2>> **<<2*3=4>>** <<4*3=12>> #### 12
***t*=4**	<<2/2=1>> <<2+1=3>> #### 3	<<43*3=129>> <<23*4=92>> <<129+92=221>> #### 221	<<32/16=2>> <<2*2=4>> <<4*3=12>> #### 12
:	**:**	**:**	**:**
***t*=1**	<<2/2=1>> <<2+1=3>> #### 3	<<43*3=129>> <<23*4=92>> <<129+92=221>> #### 221	<<32/16=2>> <<2*2=4>> <<4*3=12>> #### 12

beginning, and then the subsequent work is for refining and improving, which is also similar to how human performs a complex task. Interestingly, in DoT, the correct thoughts may not appear in a left-to-right paradigm as in the traditional chain-of-thought process. The third case serves as compelling evidence to illustrate this distinctive nature of diffusion-of-thought and how it diverges from the chain-of-thought approach. In step 4 the model has a wrong intermediate thought `<2*3=4>` with the latter thoughts and final answer computed correctly first. In the next step, the error in the wrong intermediate thought is fixed, which suggests both prior and latter thoughts can help in the prediction of the current thought. Furthermore, from these three cases, we observed that the model tends to maintain its prediction after it considers the answer to be complete. This suggests we can further enhance the inference efficiency by incorporating mechanisms such as early exit [16], and easier tasks can get earlier exits as observed in Table 3.

5 Related Work

5.1 Diffusion Models for Text

Building upon advancements in diffusion models for image generation [21, 45], text continuous diffusion [15, 33] employs an embedding function to transform discrete text into the continuous space. Besides, discrete diffusion models [1, 23] directly introduce discrete noise to accommodate the discrete nature of texts, demonstrating significant potential [37, 65]. Numerous studies have shown that diffusion models can efficiently generate diverse texts [11, 14], and achieve competitive performance in various sequence-to-sequence NLP tasks, including machine translation [60, 61], summarization [62], code generation [44], and style transfer [24]. In this work, we explore diffusion model for mathematical reasoning tasks.

5.2 Pre-train and fine-tune Diffusion LMs

The pre-training and fine-tuning paradigm, while a familiar concept in NLP before the era of prompting methods [36], remains relatively under-explored for diffusion language models. Prior efforts include initializing diffusion models with pre-trained masked language models such as BERT [20] and RoBERTa [66] and XLM-RoBERTa [59]. GENIE [35] adopts paragraph denoising to train encoder-decoder models, proving beneficial for summarization tasks. Plaid [18] and SEDD [37] are pioneers in pre-train diffusion language models from scratch, attaining comparative or better perplexity scores over GPT-2 [4]. To the best of our knowledge, we are the first to explore the fine-tuning of a pre-trained diffusion language model for reasoning tasks.

5.3 Reasoning Paradigms

Large language models usually excel in performing system-1 [47] tasks that are processed quickly and intuitively by humans but struggle in system-2 tasks, which require deliberate thinking [4, 48, 54]. The chain-of-thought reasoning paradigm [31, 41, 54] has been widely employed to elicit reasoning